CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the May/June 2014 series

0620 CHEMISTRY

0620/23 Paper 2 (Core Theory), maximum raw mark 80

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1 **(a)** **(i)** copper sulfate / $CuSO_4$ [1]

(ii) calcium oxide / CaO [1]

(iii) hydrogen chloride / HC*l* [1]

(iv) potassium bromide / KBr [1]

(v) aluminium oxide / Al_2O_3 [1]

(vi) copper sulfate / $CuSO_4$ [1]

(b) chemically; different; fixed; [3]
(1 mark each)

[Total: 9]

2 **(a)** hydrochloric (acid) / HC*l* [1]
calcium hydroxide / calcium oxide [1]

(b) ⇌ [1]

$6H_2O$ on right [1]

(c) in tube A the calcium chloride absorbs the water vapour; [1]

In tube B there is both water and air / there is water (vapour) in the air; [1]

(d) 2nd box down ticked (oxidation state of iron) [1]

(e) **(i)** magnesium < zinc < iron < lead [2]
1 mark if one pair reversed / lead > iron > zinc > magnesium

(ii) oxygen removed from the copper oxide / it loses oxygen / hydrogen gains oxygen; [1]

[Total: 10]

3 **(a)** **(i)** carrots; potatoes; [1]

(ii) (pH) 7; [1]

(b) **(i)** Any **two** from: [2]
- plants won't grow if (conditions too) acid
- to raise the pH / to make the soil less acidic / lime is alkaline / lime has high pH;
- to neutralise (the soil) / neutralisation;

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(ii) lime is alkaline / lime is a base / lime reacts with ammonium salts; [1]

ammonia produced; [1]

(ammonia) escapes (into air) / (ammonia) is a gas; [1]

(c) (i) Any **two** from: [2]
- increases;
- up to pH 7.5 / up to quoted values between pH 7 and 8;
- then levels off / evens out / then stays at the same pH

(ii) pH 9.5 / between 9 and 10 [1]

[Total: 10]

4 (a) (i) capillary tube / very narrow tube; [1]

(ii) ink would undergo chromatography / ink would run up the paper / ink masks the results / ink would smear / ink mixes with spot ORA for pencil / lead [1]

(iii) B [1]

(iv) A [1]

(v) C [1]

(b) (i) 4 [1]

(ii) 212; [2]
For 1 mark one row correct e.g.
H = 12 × 1 = 12
N = 4 × 14 = 56

(c) (i) idea of substance formed by (addition of) monomers or simple units / idea of many monomers or simple units (joined); [1]

(ii) poly(ethene) / polyethene; [1]

[Total: 10]

5 (a) (i) increases as number of (carbon) atoms increase / both increase at the same time / proportional / more carbon the higher the boiling point; [1]

(ii) boiling point **allow:** between 130 and 150 °C; [1]
(actual = 141)

Density **allow:** between 0.80 and 1.00; [1]
(actual = 0.96)

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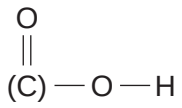
(iii) liquid because melting point below room temperature and boiling point above room temperature / room temperature is between melting and boiling point; [1]

(b)



[1]

(c) (i) burette; [1]

(ii) sodium hydroxide; [1]

(iii) indicator in flask / reference to indicator; [1]

run liquid from burette (until indicator changes colour); [1]

[Total: 9]

6 (a) $PbBr_2$ / $Pb^{2+}2Br^-$ [1]

(b) (i) to melt the lead bromide / to allow ions to move; [1]

(ii) graphite; [1]

(iii) anode: bromine and cathode: lead; [1]
(both required)

(c) (i) A; [1]

(ii) (anode): decreases in size / becomes eroded; [1]

cathode: increases in size; [1]

(iii) 134; [2]

[Total: 9]

7 (a) (i) Any **four** suitable differences e.g.: [4]

- no noble gases / only 7 (standard) Groups ORA;
- hydrogen / H in same column as Li ORA;
- some elements missing / named element missing / empty spaces ORA
- groups are horizontal rather than vertical / reference to groups or periods being different ORA
- not ordered according to atomic number / no proton numbers
- Zn put in same group as Be and Mg ORA

(ii) any **two** from: [1]
fluorine, chlorine, bromine, oxygen , nitrogen , hydrogen

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(b) any **three** from: [3]
- melting points / boiling points;
- density;
- catalytic activity;
- strength
- hardness
- electrical conductivity / heat conductivity
- malleability / ductility

(c) 2 (Cl_2); [1]
CO_2 (on right); [1]

(d) to prevent sodium reacting with air / to stop the Ti reacting with the air / to exclude air / to stop the hydrolysis of the titanium oxide / to exclude water (vapour); [1]

because argon is inert / unreactive / inactive / does not react; [1]

[Total: 12]

8 **(a)** 3rd box down ticked (giant ionic); [1]

(b) add barium chloride / barium nitrate; [1]

white precipitate; [1]
(both required)
note: second mark dependent on correct reagent

(c) Any **five** from: [5]
- condenser
- connected to flask
- mixture in flask
- idea of heating the solution / boil the solution
- water has lower boiling point than sodium sulfate / sodium sulfate is solid and water is liquid (at rtp)
- on heating water boils more easily / forms vapour more
- easily / water boils first / water will evaporate (not sodium sulfate)
- steam / water vapour goes to top of the flask and into condenser
- water vapour gets into condenser
- sodium sulfate does not turn to gas
- sodium sulfate remains in flask / sodium sulfate is left
- water vapour / steam goes to liquid in condenser
- water collected in receiver

(d) turns pink; [1]

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(e) filtered; [1]
chlorine added / chlorination; [1]
allow: other stages e.g. sedimentation / flocculation (use of iron chloride / aluminium sulfate etc.) / treatment with sulfur dioxide

[Total: 11]

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